

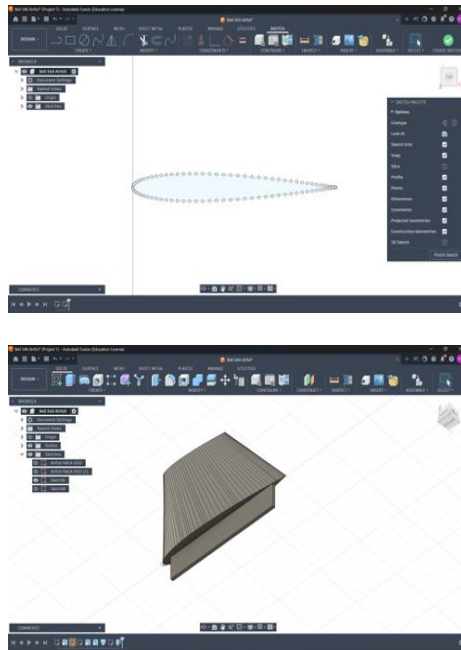
Engineering Design Review:

4-inch High-Powered Rocket Fin Optimization

CAD Modeling, Aerodynamic Analysis, and Stability Margin Optimization

Design Overview & CAD

- **Profile Selection:**
Generated NACA 0012 symmetric airfoil (45 coordinate points per side) using Autodesk Fusion Airfoil Generator.
- **Geometry Generation:** Designed root (150 mm) and tip (75 mm) chord sketches and utilized loft tool to create a smooth body.
- **Integration:** Modeled a custom 2-point rectangular mounting tab for through-the-wall attachment.

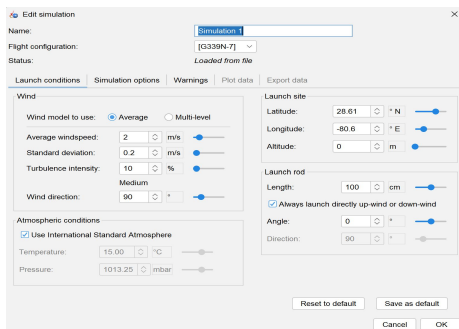


Stability & Mass Properties



- **Airframe Specs:** 122 cm body tube (10.2 cm OD, 9.91 cm ID) with a 45 cm Ogive nose (10 cm long shoulder).
- **Assembly:** 3 custom tapered fins with through-the-wall tabs mounted 3 cm from body edge.
- **Propulsion & Stability:** Configured AeroTech G339N motor. Initial simulation yielded a stability of 0.741 calibers which is too low.
- **Mass Balance Optimization:** Resolved stability issue by overriding nose cone mass from 34.6 g to 106 g (+71.4 g ballast), achieving a 1.30 caliber stability.

Flight Simulation & Aerodynamics



- **Launch Conditions:**
The default settings were used for the launch including a 2 m/s average windspeed (0.2 m/s standard deviation), and a 10% turbulence intensity.

Optimum delay	Max. velocity	Max. accelera...	Time to apogee	Flight time	Ground hit vel...
6.55 s	105 m/s	361 m/s²	6.91 s	31.5 s	12.1 m/s

- **Recovery:** A 36-in diameter parachute (CF-36, 68.9 g) was added to decrease the ground hit velocity from 12.1 m/s to 5.98 m/s. This shifted the CG forward, increasing the stability to 1.70 cal.
- **Optimum Delay:** The rocket had a 6.55 second optimum delay, so the G339N motor was adjusted to a 7 second ejection charge delay.

Optimum delay	Max. velocity	Max. accelera...	Time to apogee	Flight time	Ground hit vel...
6.47 s	98.7 m/s	338 m/s²	6.74 s	55.2 s	5.98 m/s

Aerodynamic Load & Structural Analysis

- **Dynamic Pressure Calculation:**
Calculate peak dynamic pressure (q) at 98.7 m/s.

$$q = \frac{1}{2} \rho v^2$$

Using sea-level air density ($\rho=1.225 \text{ kg/m}^3$)

$$q = \frac{1}{2} \times (1.225) \times (98.7)^2 = 5.97 \text{ kPa}$$

- Evaluated max dynamic pressure ($q=5.97 \text{ kPa}$) at 98.7 m/s (Mach 0.288). The custom NACA 0012 profile provides reduced drag and higher bending resistance compared to flat plate fins.